Use Case: Select Piece

Iteration: 1

Primary Actor: Player

Goal in Context: Allows the player to select a piece to move on their turn.

Preconditions: The game must be in progress, and it must be the player's turn.

Trigger: The player clicks on or selects a piece on the board.

Scenario:

1. The player selects a piece by clicking or entering its coordinates
2. The system highlights or marks the selected piece.
3. The system verifies that the selected piece belongs to the current player.

Post Conditions: The piece is highlighted, and the player can now select a destination.

Exceptions:

- The selected piece does not belong to the player, and an error message is shown.

Priority: High priority, as it is essential for gameplay.

When Available: Within its 1st iteration.

Frequency of Use: Multiple times per player turn.

Channel to Actor: Player interacts directly with the board.

Secondary Actors: N/A

Channel to Secondary Actors: N/A

Open Issues: None

Use Case: Move Piece on click

Iteration: 1
Primary Actor: Player

Goal in Context: Allows the player to move a selected piece to a new position on the board.

Preconditions: A piece has been selected by the player, and it is a valid piece to move.

Trigger: The player clicks on or selects a destination square on the board.

Scenario:

1. The player selects a destination for the selected piece.
2. The system checks if the move is valid according to the rules of Checkers. ○ If valid, the piece is moved to the new position.
3. The board is updated to show the new position of the piece.

Post Conditions: The piece has been successfully moved to the new position.

Exceptions:

- The move is invalid, and an error message is displayed.

Priority: High priority, as it is essential for gameplay.
When Available: Within its 1st iteration.
Frequency of Use: Multiple times per game, depending on player moves.
Channel to Actor: Player interacts directly with the board.
Secondary Actors: N/A
Channel to Secondary Actors: N/A
Open Issues: None

Use Case: Eliminate Piece

- **Iteration**: 1
- **Primary Actor**: Player
- **Goal in Context**: Allows the system to remove an opponent’s piece when a capture move is made.
- **Preconditions**: The current player has made a valid capture move.
- **Trigger**: The player completes a capture move.
- **Scenario**:
    - The system identifies the opponent’s piece that is being captured.
    - The captured piece is removed from the board.
    - The board state is updated and displayed to both players.
- **Post Conditions**: The opponent's captured piece is removed from the board.
- **Exceptions**:
    - The system fails to remove the piece due to an error.
- **Priority**: High priority, as it impacts the game's progress and rules.
- **When Available**: Within its 1st iteration.
- **Frequency of Use**: Multiple times per game, depending on capture moves.
- **Channel to Actor**: Game Controller updates the board.
- **Secondary Actors**: System
- **Channel to Secondary Actors**: N/A
- **Open Issues**: None

Use Case: Switch Player

- **Iteration**: 1
- **Primary Actor**: Player
- **Goal in Context**: Allows the system to alternate turns between players.
- **Preconditions**: The current player has completed their turn.
- **Trigger**: The player's turn is over (either after a move or capture).
- **Scenario**:
    - ○ The system verifies that the player's turn is complete.
    - ○ The active player is switched to the other player.
    - ○ The system updates the game state to reflect the new player's turn. ○ The new player is notified that it is now their turn.
- **Post Conditions**: The turn is switched to the other player.
- **Exceptions**:
    - ○ The system fails to switch the player due to an error.
- **Priority**: High priority, as it is critical for turn-based gameplay.
- **When Available**: Within its 1st iteration.
- **Frequency of Use**: Multiple times per game as turns alternate.
- **Channel to Actor**: Game Controller handles turn management.
- **Secondary Actors**: System
- **Channel to Secondary Actors**: N/
- **Open Issues**: None

Use Case: End Game

- **Iteration**: 1
- **Primary Actor**: Player
- **Goal in Context**: Ends the game when a win/loss condition is met or if a player forfeits.
- **Preconditions**: One player has no remaining pieces or legal moves, or a player chooses to forfeit.
- **Trigger**: A win/loss condition is met or a player forfeits.
- **Scenario**:
    - ○ The system checks the board state and verifies if a win/loss condition is met. ○ The game state is updated to GAME_OVER.
    - ○ The system determines and displays the winner.
    - ○ The end-game screen is displayed with options to "Play Again" or "Exit."
- **Post Conditions**: The game ends, and the result is displayed.
- **Exceptions**:
    - ○ The system fails to end the game due to an error.
- **Priority**: High priority, as it is essential for game completion.
- **When Available**: Within its 1st iteration.
- **Frequency of Use**: Once per game.
- **Channel to Actor**: Game Controller manages end-game logic.
- **Secondary Actors**: System
- **Channel to Secondary Actors**: N/A
- **Open Issues**: None

Use Case: Create King

- **Iteration**: 1
- **Primary Actor**: Player
- **Goal in Context**: Promotes a piece to a king when it reaches the opposite end of the board.
- **Preconditions**: A piece reaches the last row on the opponent's side during a move.
- **Trigger**: The player moves a piece to the last row.
- **Scenario**:
    - ○ The system identifies that the piece has reached the opponent's last row. ○ The piece is promoted to a king, with a visual indication.
    - ○ The board state is updated to reflect the promotion.
- **Post Conditions**: The selected piece is promoted to a king with enhanced movement.
- **Exceptions**:
    - ○ The system fails to promote the piece due to an error.
- **Priority**: Medium priority, as it enhances gameplay but is not essential.
- **When Available**: Within its 1st iteration.
- **Frequency of Use**: As needed during gameplay.
- **Channel to Actor**: Game Controller handles promotion.
- **Secondary Actors**: System
- **Channel to Secondary Actors**: N/A
- **Open Issues**: None

Use Case: Play Again Button

- **Iteration**: 1
- **Primary Actor**: Player
- **Goal in Context**: Allows the player to restart the game with a fresh board after the game has ended.
- **Preconditions**: The game is over, and the "Play Again" button is available on the end-game screen.
- **Trigger**: The player clicks the "Play Again" button.
- **Scenario**:
    - The player selects the "Play Again" button.
    - The system resets the game state and initializes a new board.
    - The board is reinitialized, and the players are notified of the new game start

Post Conditions: A new game begins with the board reset.

- **Exceptions**:
    - The system fails to reset the board due to an error.
- **Priority**: Medium priority, as it enhances user experience.
- **When Available**: Within its 1st iteration.
- **Frequency of Use**: Once per session, if the player wants to replay.
- **Channel to Actor**: User interface for player interaction.
- **Secondary Actors**: N/A
- **Channel to Secondary Actors**: N/A
- **Open Issues**: None

Use Case: Exit

- **Iteration**: 1
- **Primary Actor**: Player
- **Goal in Context**: Allows the player to exit the game after it has ended.
- **Preconditions**: The game is over, and the "Exit" button is available.
- **Trigger**: The player clicks the "Exit" button.
- **Scenario**:
    - The player selects the "Exit" button.
    - The application closes or returns to the main menu.
- **Post Conditions**: The game session is terminated.
- **Exceptions**:
    - The application fails to exit properly.
- **Priority**: Low priority, as it provides a way to end the session.
- **When Available**: Within its 1st iteration.
- **Frequency of Use**: Once per session, if the player wants to end the game.
- **Channel to Actor**: User interface for player interaction.
- **Secondary Actors**: N/A
- **Channel to Secondary Actors**: N/A
- **Open Issues**: None